x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i-\bar{x}) = 0$	$\sum (x_i-\bar{x})^2 = 10$	$\sum (y_i-\bar{y}) = 0$	$\sum (x_i-\bar{x})(y_i-\bar{y}) = 8$

(a) The complete table is above.

$$\bar{x} = [3+2+1+(-1)+0] \times \frac{1}{5} = 1$$

$$\bar{y} = (4+2+3+1+0) \times \frac{1}{5} = 2$$

#

(b) Calculate b_1 and b_2 and state their interpretation.

$$b_2 = \frac{\sum (x_i-\bar{x})(y_i-\bar{y})}{\sum (x_i-\bar{x})^2} = \frac{8}{10} = 0.8$$

$$b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8 \cdot 1 = 1.2$$

b_1為迴歸式 β_1 之估計值，亦為迴歸線之截距，當自變數 $x_i = 0$ 時 $\hat{y}_i$ 的估計值為 $b_1 = 1.2$ ；b_2 為迴歸式 β_2 之估計值，亦為迴歸線之斜率，當自變數 x_i 上升(下降)一單位時，$\hat{y}$ 的估計值會上升(下降) $b_2 = 0.8$ 單位 *

(c) Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$ and show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}$ and $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$

$$\sum_{i=1}^{5} x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 9 + 4 + 1 + 1 + 0 = 15$$

$$\sum_{i=1}^{5} x_i y_i = 3\times4 + 2\times2 + 1\times3 + (-1)\times1 + 0\times0 = 12 + 4 + 3 + (-1) + 0 = 18$$

$$\sum (x_i - \bar{x})^2 = 10 = \sum x_i^2 - N\bar{x} = 15 - 5\times1 = 10$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8 = \sum x_i y_i - N\bar{x}\bar{y} = 18 - 5\times1\times2 = 8$$

QED

(d) According to (b), compute the fitted value of y, and complete the remainder of the table. Put the sums in the last row.

$$\hat{y}_i = 1.2 + 0.8x_i$$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$ $(\bar{x}=1)$	$\sum y_i = 10$ $(\bar{y}=2)$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

計算 sample variance of y, $s_y^2 = \sum_{i=1}^{N} (y_i - \bar{y})^2 \times \frac{1}{N-1}$, sample variance of x, $s_x^2 = \sum_{i=1}^{N} (x_i - \bar{x})^2 \times \frac{1}{N-1}$. sample covariance $s_{xy} = \sum_{i=1}^{N} (y_i - \bar{y})(x_i - \bar{x}) \times \frac{1}{N-1}$ sample correlation $r_{xy} = \frac{s_{xy}}{s_x s_y}$. the coefficient of variation of x, $CV_x = 100 \times \frac{s_x}{\bar{x}}$, what is the median, 50th percentile of x?

$$s_y^2 = (2^2 + 0^2 + 1^2 + (-1)^2 + (-2)^2) \times \frac{1}{5-1} = 10 \times \frac{1}{4} = 2.5$$

$$s_x^2 = (2^2 + 1^2 + 0^2 + (-2)^2 + (-1)^2) \times \frac{1}{5-1} = 10 \times \frac{1}{4} = 2.5$$

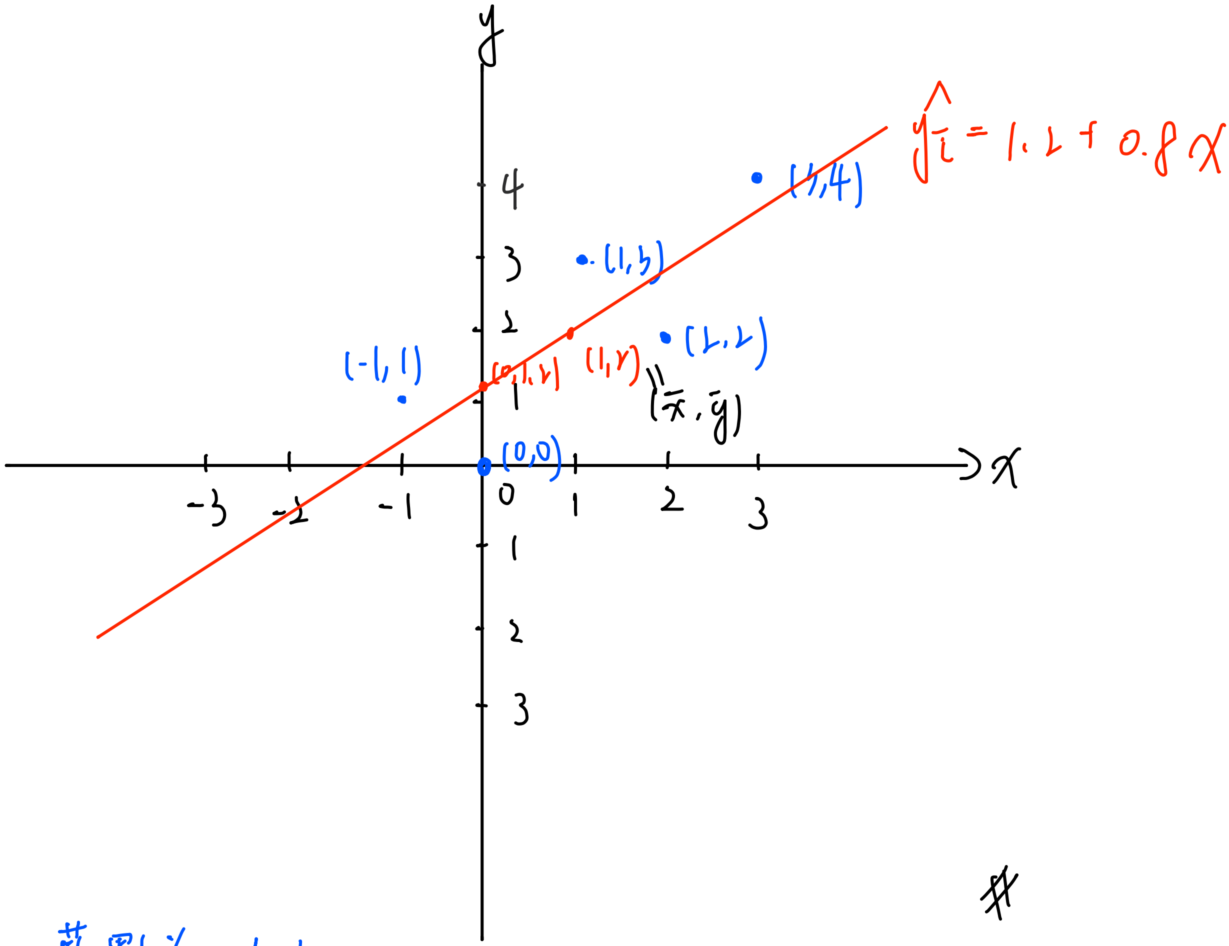
$$s_{xy} = (4 + 0 + 0 + 2 + 2) \times \frac{1}{5-1} = 8 \times \frac{1}{4} = 2$$

$$r_{xy} = \frac{2}{\sqrt{2.5} \times \sqrt{2.5}} = \frac{2}{2.5} = 0.8$$

$$CV_x = 100 \times \frac{\sqrt{2.5}}{1} = 100 \times \sqrt{2.5} \fallingdotseq 158.113883$$

x 的 median 以及 50th percentile 為同一個數，為第3大的數，故所求 $= 1$ #

(e)



#

藍點為 data points

fitted regression line $\hat{y}_i = 1.2 + 0.8x_i$

$\Rightarrow$ 過 $(0, 1.2)$、$(1, 2)$ 兩點

(f) $(\bar{x}, \bar{y}) = (1, 2)$，由 (e) 可知此點正好在 fitted regression line上 #

(g) $\bar{y} = 2 = b_1 + b_2\bar{x} = 1.2 + 0.8 \times 1 = 2$ #

(h) $\bar{\hat{y}} = (3.6 + 2.8 + 2 + 0.4 + 1.2) \times \frac{1}{5} = 2 = \bar{y}$ #

(i) $\hat{\sigma}^2 = \frac{\sum(y_i - b_1 - b_2 x_i)^2}{N-2} = \frac{\sum(y_i - 1.2 - 0.8 x_i)}{N-2}$

$= [(4-1.2-0.8\times3)^2 + (2-1.2-0.8\times2)^2 + (3-1.2-0.8\times1)^2 + (1-1.2-0.8\times(-1))^2 + (0-1.2-0.8\times0)^2] \times \frac{1}{5-2} = (0.16+0.64+1+0.36+1.44)\times\frac{1}{3}$

$= 3.6\times\frac{1}{3} = 1.2$ #

(j) $\widehat{Var}(b_2|x)$ & $se(b_2)$

$\widehat{Var}(b_2|x) = \frac{\hat{\sigma}^2}{\sum(x_i-\bar{x})^2} = \frac{1.2}{10} = 0.12$

$se(b_2) = \sqrt{\widehat{Var}(b_2)} = \sqrt{0.12} \doteqdot 0.34641$ #

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